

QIUYUE (SHIRLEY) XUE

Assistant Professor of Computer Science, Purdue University
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WORK AND EDUCATION

Purdue University Assistant Professor, Computer Science	<i>2026.01 - present</i>
University of Washington Ph.D., Computer Science & Engineering Advisors: Shwetak Patel, Vikram Iyer	<i>2019.09 - 2025.08</i>
Georgia Institute of Technology M.S., Computer Science Advisors: Gregory Abowd, Thad Starner	<i>2017.08 - 2019.06</i>
Peking University, China B.S., Computer Science B.S., Electrical Engineering	<i>2013.09 - 2017.07</i>

PROFESSIONAL EXPERIENCE

Visiting faculty researcher, Google XR Leading research on mobile sensing R&D on Google Glasses . Host: Ishan Chatterjee	Sep 2025 - Dec 2025
Research intern, Microsoft Research Research for Industry & Networking Research group. Led research on ultra-low-power sensing for the food supply chain , optimizing energy efficiency for real-world deployments. Mentor: Vaishnavi Ranganathan, Bodhi Priyantha	Jun 2022 - Oct 2022
Research intern, Apple AI/ML Led research on ultra-efficient real-time wireless sensing system for natural input interactions , enabling seamless user experiences. Mentor: Saman Naderiparizi	Jun 2021 - Oct 2021
Student researcher, Google Health Research and Innovation Led research on smart home- and smartphone-based sonar for breathing and activity monitoring . Developed deep learning models to interpret sonar signals for contactless health tracking. Mentors: D. Shin, Mark Malhotra, Anupam Pathak	Jun 2020 - Mar 2021
ML SDE intern, Bloomberg ML team Developed a sentiment analysis model for earnings calls transcripts , using NLP techniques for extracting insights and improving financial text classification. Mentor: Karan Uppal, Temma Choji	May 2018 - Aug 2018

PUBLICATIONS

1. Qiuyue (Shirley) Xue, Dilini Nissanka, Tammy Yan, Ruiqing Wang, Shwetak Patel, Vikram Iyer. **PPG Earring: Smart Wireless Earring for Heart Health Monitoring**. In Proceedings of the 2025 CHI conference on human factors in computing systems, pp. 1-16. 2025.
2. Qiuyue (Shirley) Xue, Eric Martin, Jiaqing Liu, Ruiqing Wang, Antonio Glenn, Richard Li, Vikram Iyer, Shwetak Patel. **ECG Necklace: Low-power Wireless Necklace for Continuous ECG Monitoring**. In Proceedings of the 2025 CHI conference on human factors in computing systems, pp. 1-14. 2025.
3. Qiuyue (Shirley) Xue, Yujia Liu, Vikram Iyer, and Shwetak Patel. **Design of Thermal Earring: A Low-Power Wireless Earring for Longitudinal Temperature Monitoring**. In Companion of the 2024 on ACM International Joint Conference on Pervasive and Ubiquitous Computing, pp. 366-370. 2024.
4. Qiuyue (Shirley) Xue*, Yujia Liu*, Joseph Breda, Mastafa Springston, Vikram Iyer, and Shwetak Patel. **Thermal earring: Low-power wireless earring for longitudinal earlobe temperature sensing**. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 7(4):1-28, 2024
5. Qiuyue (Shirley) Xue, D Shin, Anupam Pathak, Jake Garrison, Jonathan Hsu, Mark Malhotra, and Shwetak Patel. **Luckychirp: Opportunistic respiration sensing using cascaded sonar on commodity devices**. In 2022 IEEE International Conference on Pervasive Computing and Communications (PerCom), pages 164-171. IEEE, 2022
6. Vikram Iyer*, Maruchi Kim*, Qiuyue (Shirley) Xue* (co-first author), Anran Wang, Shyamnath Gollakota. **Airdropping sensor networks from drones and insects**. In Proceedings of the 26th Annual International Conference on Mobile Computing and Networking, pages 1-14, 2020
7. Anandghan Waghmare, Qiuyue (Shirley) Xue, Dingtian Zhang, Yuhui Zhao, Shivan Mittal, Nivedita Arora, Ceara Byrne, Thad Starner, and Gregory D Abowd. **Ubiquitous: Self sustaining ubiquitous touch interfaces**. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 4(1):1-22, 2020
8. D Shin and Qiuyue Shirley Xue. **Optical accessory to add touch capability to a non-touchscreen device**. 2020
9. Rocko Graziano, David Benton, Sarthak Wahal, Qiuyue (Shirley) Xue, P Tim Miller, Nick Larsen, Diego Vacanti, Pepper Miller, Khushhall Chandra Mahajan, Deepak Srikanth, et al. **Jack watson: Addressing contract cheating at scale in online computer science education**. In Proceedings of the sixth (2019) ACM conference on learning@ scale, pages 1-4, 2019
10. Nivedita Arora, Qiuyue (Shirley) Xue, Dhruva Bansal, Peter McAughan, Ryan Bahr, Diego Osorio, Xiaomeng Ma, Alanson P Sample, Thad E Starner, and Gregory D Abowd. **Surface++ a scalable and self-sustainable wireless sound sensing surface** (poster). In Proceedings of the 17th Annual International Conference on Mobile Systems, Applications, and Services, pages 543-544, 2019
11. Cheng Zhang, Qiuyue (Shirley) Xue, Anandghan Waghmare, Ruichen Meng, Sumeet Jain, Yizeng Han, Xinyu Li, Kenneth Cunefare, Thomas Ploetz, Thad Starner, et al. **Fingerping: Recognizing fine-grained hand poses using active acoustic on-body sensing**. In Proceedings of the 2018 CHI conference on human factors in computing systems, pages 1-10, 2018
12. Cheng Zhang, Qiuyue (Shirley) Xue, Anandghan Waghmare, Sumeet Jain, Yiming Pu, Sinan Hersek, Kent Lyons, Kenneth A Cunefare, Omer T Inan, and Gregory D Abowd. **Soundtrak: Continuous 3d tracking of a finger using active acoustics**. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 1(2):1-25, 2017

13. Cheng Zhang, Sinan Hersek, Yiming Pu, Danrui Sun, Qiuyue (Shirley) Xue, Thad E Starner, Gregory D Abowd, and Omer T Inan. **Bioacoustics-based human-body-mediated communication**. Computer, 50(2):36–46, 2017

PATENTS

Earring and earring systems including PPG sensors for heart health monitoring

Qiuyue (Shirley) Xue, Dilini Nissanka, Shwetak N. Patel, Vikram Iyer US patent 63/778,756 filed

ECG devices and systems including examples of necklaces

Qiuyue (Shirley) Xue, Eric Martin, Richard Li, Vikram Iyer, Shwetak N. Patel US patent 63/778,958 filed

Wearable devices for detecting events using external ear and ambient temperature, including examples of earrings

Qiuyue (Shirley) Xue, Yujia Liu, Vikram Iyer, Shwetak Patel US Patent 18/932,410

Opportunistic sonar monitoring of vital signs

Donggeek Shin, Qiuyue (Shirley) Xue, Shwetak Patel US Patent 17/360,999

Thin and flexible self-powered vibration transducer employing triboelectric nanogeneration

Nivedita Arora, Gregory D Abowd, Mohit Gupta, Diego Osorio, Seyedeh Fereshteh Shahmiri, Thad Eugene Starner, Yi-Cheng Wang, Zhengjun Wang, Zhong Lin Wang, Steven L Zhang, Peter McAughan, Qiuyue (Shirley) Xue, Dhruva Bansal, Ryan Bahr, Emmanouil Tentzeris US Patent 10,932,063

Systems, methods and devices for gesture recognition

Cheng Zhang, Gregory D. Abowd, Omer Inan, Pranav Kundra, Thomas Ploetz, Yiming Pu, Thad Eugene Starner, Anandghan Waghmare, Xiaoxuan Wang, Kenneth A. Cunnefare, Qiuyue (Shirley) Xue US Patent US Patent 11,762,474

TEACHING AND MENTORSHIP

Classes (teaching assistant):

University of Washington CSE/ECE 475: Embedded Capstone class, undergrad and grad level (2022 Fall, 2023 Fall)

Georgia Tech CS 6001: Artificial Intelligence, grad level (2019 Spring)

Mentored Students

Nancy (Yujia) Liu (ECE MS, now PhD student at UCSD PhD)

Jiaqing Liu (ECE MS)

Tammy Yan (HCDE MS)

Ruiqing Wang (UW Global Innovation Exchange Program, MS)

Yuhua Nie (ECE MS)

Dilini Nissanka (CSE undergrad)

Eric Martin (ECE undergrad)

Chetana Iyer (CSE undergrad)

CSNext program (for underrepresented communities): Sravya Nagalakunta, Yvonne Jose, Thanh Tong, Deepansha Singh, Ying Zhu, Wanting Mao, Wanting Mao.

AWARDS & FUNDING

University of Washington Innovation Gap Fund award (for smart earring), \$10,000, 2023

Samsung Global Research Outreach Grant Proposal, submitted, 2024

Heidelberg Laureate Forum young researcher, 2020-2022

Best poster award, Mobisys 2019

Academic Excellence Awards, Peking University, 2015

SELECTED MEDIA COVERAGE

University of Washington front page; King 5 Seattle TV; KOMO News; GeekWire; DesignBoom; FemTech Insider, etc.: “UW-developed smart earrings can monitor a person’s temperature”

ACADEMIC SERVICE

Student volunteer: UIST 2022, Ubicomp 2017

Paper reviewer: IMWUT/Ubicomp, CHI, UIST, Journal of open hardware